Network Sniffer

A **sniffer** is a tool or software designed to capture and analyze network traffic. It can intercept data packets traveling over a network and inspect their contents. Network sniffers are often used by IT professionals for legitimate purposes, such as troubleshooting, monitoring network performance, or ensuring security. However, sniffers can also be misused for malicious purposes, such as eavesdropping on sensitive data like passwords or credit card numbers.

Types of Sniffers:

1. **Hardware Sniffers**: Physical devices that connect to a network to monitor and capture traffic.
2. **Software Sniffers**: Programs like Wireshark or tcpdump used for capturing and analyzing network data.

Common Uses:

- **Network Troubleshooting**: Diagnosing issues in data transmission.
- **Security Analysis**: Identifying potential vulnerabilities or detecting unauthorized activities.
- **Protocol Analysis**: Studying how protocols behave on the network.
- **Eavesdropping**: Capturing sensitive data (when misused).

Ethical Considerations:

- **Authorized Use**: Sniffers should only be used on networks where you have explicit permission.
- **Confidentiality**: Ensure data captured is handled securely and ethically.

Would you like to know about specific sniffer tools or how to use them?